

CHEM 101 Topic 5: Periodic Properties of Elements — Problem Set

Instructions: Complete the following problems using the analysis method practiced in your guided assignment. Show all your work, including your reasoning for trends and your electron dot diagrams.

Section 1: Organizing the Periodic Table

Problem 1: Compare a period to a group. Which one contains elements with similar chemical properties? Explain why.

Problem 2: If an element is classified as a "metalloid," what does that tell you about its physical and chemical properties?

Problem 3: Why was Moseley's shift to atomic number more accurate than Mendeleev's use of atomic mass?

Section 2: Periodic Trends

Problem 4: Between Sodium (Na) and Fluorine (F), which atom has a larger atomic radius? Justify your answer using periodic trends.

Problem 5: Between Lithium (Li) and Cesium (Cs), which element has a lower ionization energy? Explain why.

Problem 6: Compare the size of a neutral Magnesium (Mg) atom to its ion (Mg^{2+}). Which is smaller and why?

Section 3: Ions and Charges

Problem 7: Predict the ionic charge for the following elements based on their group position: a) Potassium (K); b) Sulfur (S); c) Barium (Ba); d) Bromine (Br).

Problem 8: Why do the Noble Gases (Group 18) rarely form ions?

Section 4: Electron Dot (Lewis) Notation

Problem 9: Draw the electron dot diagrams for the following elements: a) Beryllium (Be); b) Nitrogen (N); c) Oxygen (O).

Problem 10: Phosphorus (P) is in Group 15. How many valence electrons does it have, and how many more does it need to complete its outer shell?